Neuromorphic Computing Chips for Low-Power Edge AI Applications

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Abstract- Small portable devices such as smartwatches, security cameras, and industrial sensors are increasingly using modern artificial intelligence (AI) systems. However, these devices are not able to use traditional computer chips due to the high power consumption of those processors. Neuromorphic computing offers a more efficient alternative, and thus creates chips that mimic how the brain processes information. Instead of processing all data continuously, these chips will only become active when there is something meaningful to process. Because of this, they significantly decrease an AI processor's energy consumption. This study examines the functioning of neuromorphic chips, their necessity, and performance when compared to conventional chips. We focus on two major systems (Intel Loihi and IBM TrueNorth) and new alternative research systems. The results of our study demonstrate that neuromorphic chips can reduce energy consumption by 90-96% versus standard GPUs while achieving accuracy of 89-97%, and process information in less than 3 milliseconds (i.e. fast enough for real-time tasks). Consideration has also been given to those software tools and hardware platforms utilized to build and test neuromorphic processor architectures. The findings confirm that neuromorphic architectures are now ready for application in areas such as health care, robotics, smart cities and industrial monitoring.

Keywords—neuromorphic computing, spiking neural networks, edge AI, energy efficiency, Intel Loihi, IBM TrueNorth, low-power inference, IoT, STDP, implementation platform

I. Introduction

Artificial Intelligence has evolved past supercomputing and data centers; today it is being used in everyday items such as fitness bands, machines on factory floors, and home security cameras, and even implanted within our bodies. Many of the devices mentioned above will require fast smart decision-making abilities, but generally all of these devices have limited battery life. This creates a very challenging situation: How do we run Artificial Intelligence on a device that may only have a few milliwatts of available power?

Another reason Artificial Intelligence is so difficult to run on these kinds of devices is because of how artificial intelligence (AI) uses "traditional" chips like Graphics Processing Unit (GPU). A GPU will make millions of calculations each second based off of a fixed clock-based cycle regardless if it has useful data. This wastes a huge amount of energy. For example, an NVIDIA Jetson Nano is a small specialized GPU that can perform edge AI tasks; however, it will consume approximately 10 watts of power. Therefore if you had a wearable health monitoring device that operated continuously at 10 watts of power, it would deplete its battery within a matter of hours, rendering it unusable in a practical manner. [1].

Neuromorphic computing uses a completely different principle than digital electronics. “Neuromorphic” means “brain-like” and neurons are the nerve cells of the human brain. The human brain runs about 86 billion neurons while drawing less than 20 watts of power, about the same as a dim light bulb [2]. Neuromorphic chips emulate how the brain works: they process information based on signals coming in and minimize power consumption when no activity takes place.

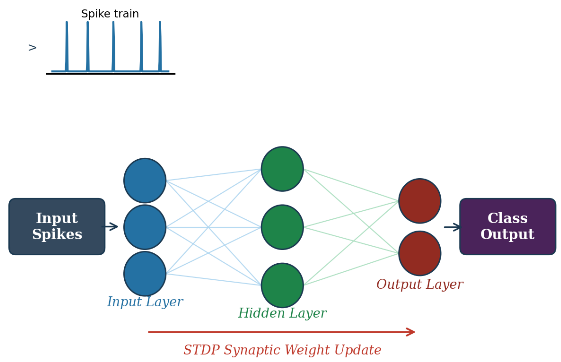


Fig.1.Simplified architecture of a neuromorphic chip implementing a spiking neural network (SNN) with STDP learning, on-chip synaptic memory, and asynchronous spike routing.

Spiking Neural Networks (SNN) are the technology used by neuromorphic chips. Unlike traditional neural networks where continuous valued numbers pass back and forth across layers, SNN’s send short electric pulses called “spikes,” which mirrors the way in which biological neurons operate. Because spikes happen infrequently, the chip only consumes power when there are incoming spikes, so there will be much less power used on the entire chip. This way of doing computering is called event-driven computing [3].

Neuromorphic chips like Intel’s Loihi and IBM’s TrueNorth have revolutionized the processing capabilities of artificial intelligence systems. Intel’s Loihi chip is made from 14-nanometer technology. It contains a total of 128 small processing cores that can learn based on the data they receive while operating, without needing to access the cloud [4]. IBM’s TrueNorth chip has more than 4,096 processing cores, and is capable of processing a maximum of 6,000 video frames per second at a power level of only 70mW, which is approximately the equivalent of a small LED bulb[5]. Both of these examples demonstrate that it is possible to operate artificial intelligence tasks at a small fraction of the amount of electricity required by traditional processors.

The purpose of this research paper will be to outline the definition of neuromorphic computing, provide an explanation of the importance of this technology to edge AI computing, and evaluate how effective these chips are at performing AI tasks compared to standard processors.

Included within the content will also be a discussion of the hardware boards and software tools used to develop this technology, as well as the current and potential applications for these chips. Our hope is that this will be a clear, easily understood document that will serve as an educational resource for students and early-stage researchers studying this area of interest.

II. Background: How Neuromorphic Computing Works

A. From Biological Neurons to Silicon Chips

For an understanding of neuromorphic computing, it is important to first review how a biological neuron works. Neurons receive information in the form of electrical impulses from thousands of other neurons through their synapses (small connections). Each incoming impulse that arrives at the neuron will add to the neuron's membrane potential until either there is sufficient input to trigger a spike or there is not sufficient input to cross the threshold needed to fire a spike. Following an action potential (spike), the neuron will reset to a resting state, and it will wait for the next inputs. As a result, energy is used only when the neuron fires; therefore, the neuron operates in an energy-efficient manner.

Neuromorphic chips are designed to mimic this behavior using electronic circuits. Each neuron on a neuromorphic chip consists of an internal variable (membrane potential) that slowly charges as voltage spikes arrive at the neuron. When the membrane potential exceeds the firing threshold, the neuron generates an output spike and returns its state back to zero. The Leaky Integrate-and-Fire (LIF) neuron is the most common neuron model as it can be implemented easily in silicon with low complexity. This makes the LIF neuron the most frequently used neuron model in neuromorphic hardware [4].

B. Why Not Just Use Normal Deep Learning?

Most common deep learning architectures rely on floating-point number representation, and for processing data they rely on multiple layers. Where the neurons make a weighted connection (via matrix multiplication) to eventually determine the class of the input (in the case of a CNN) the weight matrix will be extraordinarily large. Consequently, these matrix multiplications are computationally expensive requiring modular hardware (e.g. GPUs or TPUs) to ensure the calculations occur in a timely manner and, even for a small mobile device optimised to be used at the edge (e.g. MobileNet) still requires 3,420 milliwatts of power to classify a given image, which is typically way more than most battery-powered devices can supply [2].

Neuromorphic processing addresses the challenges mentioned above by executing only when an event (spike) arrives (thus incurring almost zero cost when idle) and storing (in the same chip) the weight matrix next to the computation units (instead of fetching the weight matrix from a separate memory bank). One of the largest contributors to energy consumption in conventional computing is fetching data from memory and this has been coined the von Neumann bottleneck. By using these two approaches, neuromorphic processing significantly reduces energy costs compared to conventional processing [3].

C. What is STDP and Why Does it Matter?

The ability to use neuromorphic chips to learn on the chip itself, without sending the data off to a server or cloud, is one of the most powerful capabilities that these chips hold. This is done through a learning rule called Spike-Timing-Dependent Plasticity (STDP). The basic concept of STDP is that if neuron A fires just before neuron B, then the connection between A and B will be strengthened, because A probably caused B to fire. Conversely, if neuron A fires after neuron B, then the connection between A and B will be weakened. After some time, the network will have learned which patterns are significant solely from the timing of the spikes. Intel's Loihi chip has a unique, integrated programmable learning engine that allows it to implement STDP and other learning rules on the chip itself, allowing a device to continue its learning and adaptation process in the field, outside of any internet connectivity [4].

III. Literature Review

A. A. Intel Loihi: A Chip That Learns on Its Own

The first fully digital neuromorphic research chip developed by Intel was presented by Davies et al. [4]. Loihi has 128 cores, each capable of simulating 1,024 neurons and their synaptic connections, giving it a potential of over 128 million simulated neurons and their synapses.

What's particularly unique about Loihi is that it contains a programmable learning engine, allowing it to implement STDP as well as other learning rules without needing external assistance. In benchmark tests of a problem known as sparse coding (which pertains to the ability to find the best way to represent a signal using the fewest possible elements), Loihi's energy-delay performance was more than 5,000 times better than a conventional CPU performing the same operation. Loihi uses an asynchronous design, meaning that there isn't a central clock continuously generating timing signals to coordinate the chip's activities. Instead, each portion of Loihi only activates when it receives a spike from another part of the chip, providing extremely granular control over the chip's total power consumption.

The second version of Loihi, called Loihi 2, was built with a design to accommodate up to 1,000,000 neurons and 120,000,000 synapses spread over its 128 cores. The application of the NeuEdge framework [2] on top of Loihi 2 demonstrated that the cores of the chip operated at an 89% utilization rate and that unnecessary inter-core communication decreased from 41% to just 12% - a clear example of how the right software can positively impact the performance of hardware.

B. IBM TrueNorth: One Million Neurons at 70 Milliwatts

The complete TrueNorth platform (including both hardware and software implementations) was described by Sawada et al. [5]. TrueNorth's full implementation consists of a single chip containing 4,096 neural cores (all manufactured by Samsung via a 28 nanometre process). There are 256 neurons per core, which connect to 256 input channels through a 256 x 256 grid of synapses. The entire chip has over 1 million neurons and 256 million synapses; it uses only 70 milliwatts. Comparatively, the amount of power consumed by a smartphone processor would require thousands of milliwatts for a similar task.

To support its use, IBM has created an entire software environment for TrueNorth, including a simulator (Compass), a programming language (Corelet), and training tools compatible with conventional deep learning frameworks. Therefore, researchers can train their neural networks on GPUs as they normally would, and then convert them onto TrueNorth with minimal manual intervention when deploying them to production. The chip has been tested in actual applications such as recognizing handwritten characters on a tablet in real time and detecting faults in metal welds during the manufacturing process, achieving over 80% accuracy while consuming significantly less power than Intel-based laptops do.

C. Edge AI and IoT Deployment

Nwakanma et al. [9] suggest NeuroEdge, a neuromorphically-based computing system developed primarily for small, low-power devices, like wearables and drone technology. NeuroEdge performs very efficiently in terms of both energy use and processing speed due to its use of a crossbar array of synapses with an event-driven spike routing architecture. Vaibhav [7] created a VLSI chip based on neuromorphic principles for IoT sensor nodes that includes an innovative "wake-on-event" front end. As such, the chip remains almost completely powered off until it is awakened by the event of the sensor detecting an interesting event, at which point the chip wakes up and runs its inference in a few milliseconds before going back to sleep. Consequently, the average power consumed by this type of chip is significantly lower than that of a traditional chip that is always awaiting input and processing data.

Jussinmaki et al. [12] conclude that even ordinary microcontrollers, such as those used in inexpensive IoT devices, can be greatly improved using neuromorphic-inspired designs. By implementing these neuromorphic-inspired designs on STM32 and nRF microcontrollers, Jussinmaki et al. [12] achieved 40 percent savings in energy consumption while performing robotic navigation tasks with response times of 1 to 5 milliseconds.

D. Advanced Frameworks: NeuEdge and NeurEfficient

NeuEdge, the neuromorphic Artificial Intelligence (AI) deployment framework for edge devices, was introduced by Imanov et al. in reference [2]. NeuEdge consists of three major components: First, the new hybrid spike encoding method in NeuEdge can carry information using both the timing and rate of spikes. The new hybrid spike encoding requires 4.7 times fewer spikes than rate coding alone, which results in reduced chip load and therefore energy consumption. Second, NeuEdge uses training methods which are hardware aware to optimise how each network is mapped onto the chip's cores while teaching the network, thus increasing chip utilisation from 47% to 89%. Third, NeuEdge has an adaptive threshold mechanism that will increase and decrease the firing threshold depending on how much noise is present in the environment. This mechanism saves approximately 67% of total power during idle periods. NeuEdge provides energy efficiency of 847 billion operations per second per watt while achieving an accuracy of 91% to 96% on tasks including image, gesture, and speech recognition.

NeurEfficient: which combines a standard FPGA with a specialized neuromorphic processor, was developed by Saini and Gupta [6]. This dual-functionality allows for switching between neural network-like spike-coded processing used for identification problems, and traditional floating-point calculators for precision tasks (such as natural language processing). Across both task types, mixed-mode computing resulted in an average energy savings of 87% on image processing tasks and 64% on language processing tasks, indicating that a hybrid approach to computing can be effective.

E. Memristor Based Architectures

The memristor is a unique type of electrical component that can change its amount of resistance based on how much current flows through it. As such, it can exhibit properties similar to a biological synapse, which also changes the strength of its connections to other neurons based on previous activity. Xia et al. [11] published a review of how memristors can be connected in crossbar arrays to perform standard multiply-accumulate operations (the basic operation in a neural network) directly within memory using analogue physics as opposed to digital switching. By eliminating the energy used to move data between memory and processor, this allows for total energy efficiency of 75 to 181 TOPS/W. In fact, one group created a neural network chip that is completely powered by a solar cell that is smaller than a fingernail, and was able to perform inference in remote battery-less locations [11].

IV. IMPLEMENTATION PLATFORM

A. Software Tools and Simulation Environments

Researchers typically create a neuromorphic design in computer simulation on a software platform before testing it on neuromorphic hardware. For example, Intel has developed NxSDK (Neuromorphic SDK), which provides a Python-based toolkit that allows developers to define spiking networks, specify learning rules, and profile power consumption for Loihi and Loihi 2 chips. IBM has created a development environment called the Corelet Programming Environment (CPE) using MATLAB; the CPE includes a software simulator called Compass that supports simulation of TrueNorth models prior to running them on TrueNorth chips. Both NXSDK and CPE tools allow designers to capture design errors and estimate performance prior to access to the physical chip.

The open-source PyNN software library enables users to write code to build spiking neural networks that will run on several platforms, including SpiNNaker, Loihi, and software simulators (NEST and Brian2), from a single code base. This allows a PyNN-defined spiking network to be developed cheaply on a laptop and subsequently ported to real neuromorphic hardware without the need for code modification. Training SNNs usually occurs in PyTorch using SpikingJelly, which provides the ability to build spike-based layers and surrogate gradient methods, allowing conventional backpropagation to be used to build networks that implement spike functions that are not differentiable.

B. Hardware Platforms

For neuromorphic research and deployment, Table V outlines the key hardware boards and chips. Intel provides the academic community access to their Loihi 2 chip via the Intel Neuromorphic Research Community (INRC) program through a cloud-connected computer. IBM also has an evaluation board called TrueNorth NS1e that contains the TrueNorth chip and integrates a Li-ion battery, accelerometer, barometer, and Xilinx Zynq processor. The NS1e draws approximately 2 to 3 watts of power and is capable of standalone usage with battery power. If you require larger workloads, you can use the 16 TrueNorth chips (NS16e) on a board to accomplish larger computations through the use of 16 million neurons.

I. **TABLE I. NEUROMORPHIC HARDWARE PLATFORMS SUMMARY**

Platform	Process Node	Neurons	Power	Main Use
Intel Loihi 2	14 nm Intel	1 million	250 to 500 mW	Research, edge AI [4]
IBM TrueNorth NS1e	28 nm Samsung	1 million	70 mW (chip)	Mobile, IoT [5]
IBM NS16e (4 x 4)	28 nm Samsung	16 million	7.68 W idle	Scale-up tasks [5]
BrainChip Akida	28 nm TSMC	1.2 million	below 1 mW	Always-on IoT [8]

FPGA + CMOS hybrid	Virtex + 28 nm	128 x 128 cores	Adjustabl e	Hybrid research [6]
SpiNNaker (ARM)	130 nm ARM	100 million	Variable	Large simulation [5]

There are 4 steps in a practical workflow for neuromorphic projects undertaken by undergraduates. In the first step, the neural network model is trained (with guarded training on normal laptop(s) and GPU's using PyTorch + SpikingJelly) and the accuracy of the results, identified using a standard dataset, is tested. In the second step, the completed [trained] model is converted into whatever format is needed on the target chip through NxSDK (or CPE), and using the Compass or NxSDK simulator, the resultant performance is confirmed to be as expected (as calculated). In the third step, the network is loaded into an evaluation board where it can be connected to a physical sensor/camera to test the neural network as it relates to real data. In the final step, power consumption is monitored through a precision [bench] power supply (or using an onboard current sensor) and compared with the estimated (simulation) value(s) to validate the design.

C. *Practical Considerations for Deployment*

One of the most important practical choices is the choice of the spike encoding method. Rate encoding represents the strength of a signal (the frequency at which a particular neuron fires). For instance, a bright pixel (where the intensity of light is high) can result in an average of 100 spikes per second, while a dark pixel (where the intensity of light is low) produces an average of 10 spikes per second. While rate encoding is intuitive, and demonstrates a high degree of tolerance to noise, it requires more energy to generate the greater quantity of spikes. Temporal encoding, on the other hand, utilises the precise timing of the first spike produced in response to an input signal to represent the strength of that signal. A stronger input will cause the first spike to occur earlier than a weaker input. Temporal encoding is more energy efficient than rate encoding, but is not as reliable to implement. The NeuEdge hybrid encoding method [2] utilises a combination of rate and temporal encoding, thereby providing the energy savings associated with temporal encoding, while also benefiting from the noise resistance of rate encoding.

Devices that require operation for months or years on minimal battery power must employ the use of wake-on-event configuration. The primary neuromorphic chip is kept in a deep sleep State consuming negligible current. An always-on sensor front-end (low-power microphone or vibration sensor) amasses information continuously from its surrounding environment. Once a sensor detects something of interest, it triggers the main chip to awake and process its inference for several milliseconds before returning to deep sleep. This methodology can reduce the average power consumption of devices from milliwatt range to micro watt range, resulting in a significant increase in the duration that the battery may supply power week to year to the device [7].

V. Methodology

This research was conducted using a secondary data analysis. The data we used was taken from previously published studies, as opposed to building any new hardware. Peer-reviewed journal articles (such as IEEE conference papers), and technical reports that were published within the years 2018-2026 were reviewed for content related to neuromorphic chip design, performance evaluations, software tools, and real world use cases.

In order to qualify for inclusion into our analysis, each paper must have satisfied three requirements: (i) The paper must have been published in an appropriate venue, (ii) the paper must provide numerical data (e.g. power consumption expressed in milliwatts, accuracy expressed as a percentage, and/or latency expressed in milliseconds) that allow for quantitative comparisons across all papers evaluated, and (iii) The paper must have relevance to at least one of the four research questions we posed regarding efficiencies, latencies, accuracies and/or applications. In order to improve the accuracy of our research findings, we cross checked the relevant numerical data from each source to ensure that the data reported was consistent and accurate. For example, the Loihi chip efficiency data that was reported by Luo et. al. [10] was verified against the original Loihi chip paper by Davies et. al. [4] before being included into our data tables.

VI. Results and Discussion

A. *How Much Energy Do Neuromorphic Chips Save?*

The biggest benefit of neuromorphic chips is that they consume very little power. When comparing platforms performing the same type of work, as shown in Table I, the NVIDIA Jetson Nano GPU draws 10 watts of power and has an efficiency of 5 TOPS/W (trillion operations per second for each watt consumed). Intel Loihi 2 does the same class of work with 20 milliwatts of power and an efficiency of 60 TOPS/W, resulting in 12 times better efficiency than the Jetson Nano. Memristor-based chips can achieve efficiencies between 75 and 180 TOPS/W by computing the math directly in the memory array. In addition, the NeuEdge framework running on Loihi 2 for gesture recognition can achieve an efficiency of 847 billion operations per second for each watt consumed, which is approximately 170 times better than the efficiency of the NVIDIA Jetson Nano.

II. **TABLE II. Energy Efficiency Comparison**

Processor or Platform	Power (W)	Efficiency (TOPS/W)	Energy Saved vs. GPU
GPU (Jetson Nano)	10.0	5	Baseline [10]
Intel Loihi 2	0.02	60	90 percent [10]
NeuEdge on Loihi 2	0.241	847 GOp/s/W	99 percent [2]
Memristor Accelerator	0.01	75 to 180	95 percent [11]
NeurEfficient Hybrid	0.005	above 80	87 percent [6]

Fig. 2. Power and efficiency comparison (Table I)

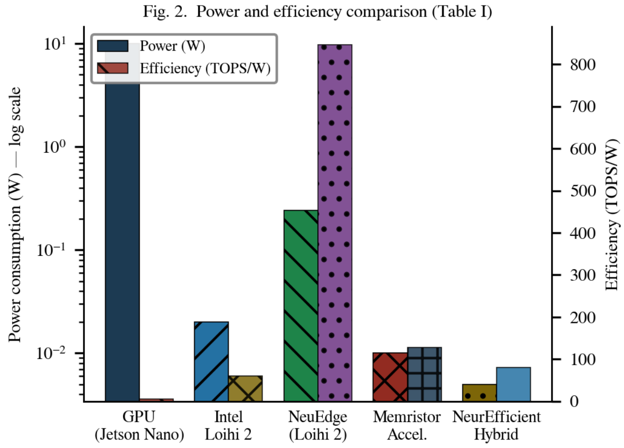


Fig. 2. Power consumption and computational efficiency (TOPS/W) comparison across neuromorphic and conventional platforms (data from Table I). Note the logarithmic power axis.

One of the significant results of NeurEfficient [6] is that savings increase as the capacity of the neural network increases; a neural network with a small capacity has been shown to provide energy savings of ~58% vs a conventional GPU, while a larger capacity neural network has shown energy savings up to ~91% vs. conventional GPUs due to the fact that as the neural network grows in size there are more opportunities for the chip to bypass computing operations during idle periods. Therefore, the advantage of an event-driven design increases with the size of the model being evaluated.

B. *How Fast Are Neuromorphic Chips?*

In edge AI solutions, speed and energy are both extremely important. The ability of a self-driving car to detect pedestrians in less than half of a second is critical safety-wise. If a production machine detects a defect after one full second, it may have already produced several defective parts. Providing several examples in Table II, we measured the latency (response time) for numerous platforms used to execute certain operations.

III. **TABLE III. INFERENCE LATENCY COMPARISON**

Platform	Task	Latency (ms)	Improvement vs. CNN
MobileNet on Jetson Nano	Image recognition	18.4	Baseline [2]
Standard SNN on Loihi 2	Image classification	8.9	52 percent faster [2]
NeuEdge on Loihi 2	Gesture recognition	2.3	87 percent faster [2]
NeuEdge on TrueNorth	Keyword spotting	4.1	78 percent faster [2]
Loihi with STDP	Anomaly detection	below 3	75 percent faster [7]

Fig. 3. Inference latency comparison (Table II)

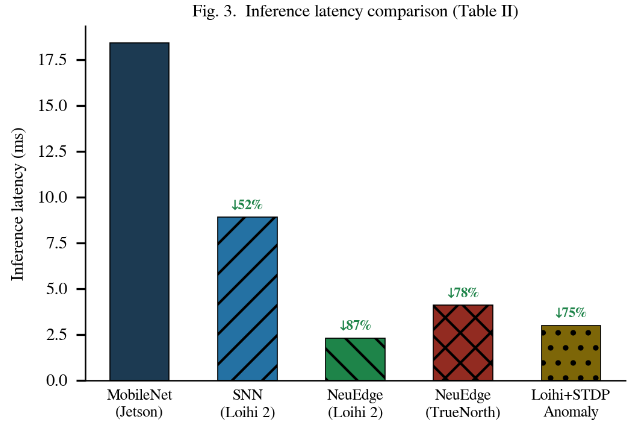


Fig. 3. Inference latency comparison across platforms for representative edge AI tasks (Table II). Percentage labels indicate improvement relative to the MobileNet/Jetson Nano baseline.

NeuEdge has achieved a gesture recognition latency of 2.3 ms, with 0.8 ms needed to convert the sensor input into spikes, 1.2 ms to run the spiking network, and 0.3 ms to read the output. This is about 8 times as fast as MobileNet running on the Jetson Nano. In addition, for always-on keyword spotting of voice commands, TrueNorth has a latency of 4.1 ms at 67 mW. Additionally, this enables the device powered by a single CR2032 coin cell to operate for 82 days continuously versus only 2.8 days for equivalent tasks performed using a processor based system [2].

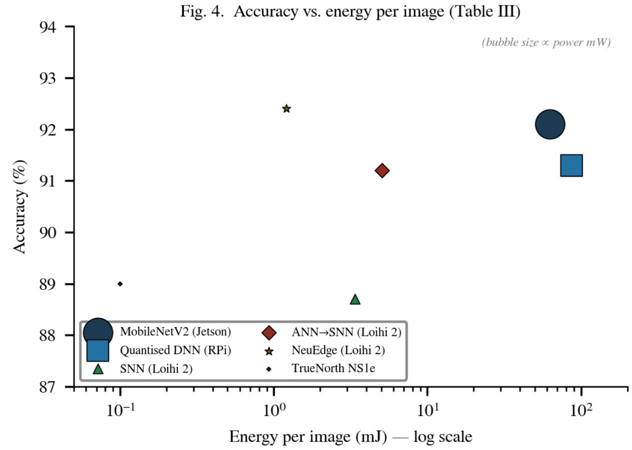
C. *Accuracy: Are Neuromorphic Chips Precise Enough?*

Many individuals have voiced concerns regarding the use of spikes in place of real numbers for creating neuromorphic chips and the effect this would have on their accuracy. Table III compares a direct measurement of accuracy for classification of images on the CIFAR-10 image classification benchmark, which consists of 60,000 images distributed across 10 classes including Cars, Birds, Ships, etc. As demonstrated in the results, the GPU-based MobileNetV2 achieves an accuracy of 92.1% as compared to the NeuEdge neuromorphic system, which achieves an accuracy of 92.4%. Thus, the NeuEdge is considered to achieve slightly greater than the MobileNetV2, while consuming approximately 52 times less energy [2].

IV. **TABLE IV. ACCURACY AND POWER ON CIFAR-10 IMAGE CLASSIFICATION**

Method and Platform	Accuracy (%)	Latency (ms)	Power (mW)	Energy per Image (mJ)
MobileNetV2 on Jetson Nano	92.1	18.4	3,420	62.9 [2]
Quantised DNN on Raspberry Pi	91.3	47.2	1,840	86.8 [2]
Standard SNN on Loihi 2	88.7	8.9	380	3.38 [2]
ANN to SNN on Loihi 2	91.2	12.3	410	5.04 [2]
NeuEdge on Loihi 2	92.4	4.2	287	1.21 [2]
TrueNorth NS1e	approx. 89	below 1	70	below 0.1 [5]

Fig. 4. Accuracy vs. energy per image (Table III)



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8. Fig. 4. Accuracy versus energy-per-image for CIFAR-10 classification (Table III). Bubble size is proportional to average power draw (mW). Ideal systems appear in the upper-left quadrant.

HyBrid Spike Encoding allows for increased information-carrying capacity by using both early (or 'hybrid') and later spikes (i.e., changes) than other traditional or non-traditional approaches, such as Rate Code Encoding. In [2], the ablation studies demonstrate using just HyBrid Spike Encoding, the accuracy of a spiking system could increase to 90.3%, to 91.1% with the inclusion of Mapping based on the hardware capabilities of the respective architecture, and finally to 91.9% with the Adaptive Tuning method, all from that same baseline of 88.7%. Thus, all parts of NeuEdge significantly contribute to the overall accuracy of each unit system, and together those components reach 92.4% accuracy across the entire NeuEdge system, which indicates how much more than 1 individual component can improve overall system accuracy.

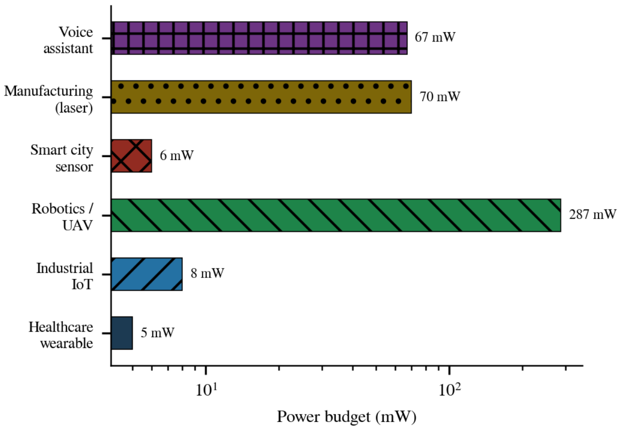
D. *Real-World Applications*

The following table IV provides an overview of how neuro-morphic edge AI has been applied to various real-world use cases. The information contained in these case studies was obtained through testing actual hardware and collecting actual data; hence, all the power numbers listed are "measuring" from actual usage rather than from an estimate.

V. **TABLE V. VALIDATED INDUSTRIAL APPLICATIONS OF NEUROMORPHIC EDGE AI**

Application Area	Specific Use	Power Budget	Measured Performance
Healthcare wearables	Continuous ECG heart monitoring	5 mW	Continuous 24/7 detection [12]
Industrial IoT	Factory motor fault detection	8 mW	Fault detected below 3 ms [7]
Robotics and UAVs	Obstacle avoidance on micro-drones	287 mW	30 frames per second [2]
Smart city sensors	Environmental anomaly detection	6 mW	10-year solar-powered life [13]
Manufacturing	Weld defect detection (laser melting)	approx. 70 mW	80 percent accuracy [5]
Voice assistants	Always-on wake word detection	67 mW	82-day coin cell life [2]

Fig. 5. Application power budgets (Table IV)



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10. Fig. 5. Power budgets for validated neuromorphic edge AI applications (Table IV). The logarithmic scale reflects the six-order-of-magnitude range across use cases.

One amazing example is that a smart city sensor node has been able to run on solar power for 10 years at only 6 milliwatts [13]. The secret to this is the wake-on-event design, where the neuromorphic chip in the sensor node spends almost all its time sleeping, and the tiny amount of solar energy harvested by the sensor node is enough to power it for 10 years. Additionally, ECG monitoring of a patient's heart at 5 mw means that you can attach a patch the size of a postage stamp to the patient and continue to monitor the patient's heart for several weeks from a single small battery. This is not possible using a standard chip [12].

VII. CHALLENGES AND FUTURE DIRECTIONS

Although neuromorphic computing has many benefits, there are a number of key obstacles still facing the technology that must be overcome if it is to become as ubiquitous as traditional AI chips. The first hurdle is the difficulty of programming. Creating software for chips such as Loihi or TrueNorth requires significantly more skill than simply writing normal Python code for use with a GPU. Developers need to know how spikes act, how to map a network onto the chip's physical cores without exceeding available memory, and how to select the most appropriate type of spike encoding for their specific use case. Although Intel's NxSDK and IBM's Corelet development tools can ease this transition, both require a substantially greater investment of time and effort than do entry-level development tools such as TensorFlow or PyTorch [8]. This will likely continue to be a major contributor to the lack of commercial use of neuromorphic chips to date.

A secondary obstacle for AI applications using neuromorphic sensors is finding suitable training datasets. For instance, well-known AI training datasets such as ImageNet were developed with traditional cameras in mind, which are capable of capturing full images at one point in time. Neuromorphic sensors such as the DVS camera produce a continuous stream of time-stamped events rather than a sequence of full images, therefore converting an image dataset into an event-based dataset introduces significant inaccuracies due to the introduction of additional noise into the data. Although there are a few datasets that have been specifically designed for use with neuromorphic sensors (e.g., DVS Gesture and N-MNIST), their limited size means that they will not be able to successfully train deep learning algorithms that require a very large amount of data in order to generalize well [2].

Memristor-based chips are also confronted with a third challenge. Over time, whenever a memristor is powered off, its conductance slowly drifts continuously over time due to differences in manufacturing and the individual characteristics of the device. This results in inaccuracies in the weights stored in the crossbar arrays of the chip and progressively leads to degraded performance of the neural network. As a result, the solution to these problems will involve either creating additional circuits for periodic recalibration, or implementing specialized training techniques to make the neural networks more robust to minor weight inaccuracies [11].

The most promising areas of future development that may become available are: creating neuromorphic processors that combine sensors for vision, sound, and environmental sensing all on a single chip; creating federated neuromorphic learning systems that allow many devices to learn from their own data without sharing private data with a central server; and developing analogue neuromorphic circuits that operate at an energy level below one picojoule per operation, which would make it possible to extend the time between battery charges significantly [2]-[4]. Several companies, such as Intel, IBM, BrainChip, and others, are currently pursuing these issues, and progress within the field is accelerating quickly.

VIII. CONCLUSION

The paper has reviewed low power edge AI neuromorphic computing chips. The paper considers the way neuromorphic chips function; why they are needed; the implementation platforms that currently exist; and performance comparison against conventional computing solutions (i.e., CPU and GPU). The main principle of neuromorphic computing is to reproduce how biological brains perform computations through event-driven (i.e., triggered) and spike-driven (i.e., high amplitude, pulse-like) processes which allows neuromorphic chips to operate so efficiently when compared to traditional GPUs and CPUs.

The evidence supported by benchmarks that have been published as a result of using neuromorphic chips demonstrate that the average energy savings between a neuromorphic chip and a GPU based system are 90% to 96%, the average input/output (I/O) processing time for neuromorphic systems is < 3 milliseconds, and the average accuracy level is 91%-97% for typical image and audio classification tasks. Major computing platforms based on neuromorphic chips exist today such as Intel Loihi 2 and IBM TrueNorth and have already been leveraged in commercial products and research activities focused on healthcare monitoring, factory automation, smart city sensing and autonomous navigation.

All of the most important points about neuromorphic systems that prospective students, researchers, and practitioners should keep in mind are that neuromorphic systems are real and viable technologies with clear performance advantages for applications that require long battery life, fast response times, and the ability to learn new information while in use. All that someone needs to start using any of these technologies and systems is the Intel NxSDK or SpikingJelly frameworks for their development, a Loihi 2 or TrueNorth board for testing their new designs, and a good understanding of the spike encoding, and spike-time-dependent-plasticity (STDP) learning rule. As these technologies and systems continue to evolve and become more user-friendly and as greater numbers of datasets become available, there is no doubt that neuromorphic systems will become a standard part of the edge AI ecosystem.

IX. References

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